

Module 7: From Political Economy to (Micro)Economics (1871–1933)

Political & Economic Thought — Master Notes · by Jakob V. Stangel

Readings: Backhouse, *The Penguin History of Economics* (2023), pp. 185–205 (the split between history & theory; the marginal revolution) · W. S. Jevons, *The Theory of Political Economy* (1871, 3rd ed. 1888), **chs I + III** · Alfred Marshall, *Principles of Economics* (1890, 8th ed. 1920), **Bk V ch. III** (the reading list says "ch. II"; the supplied text is Bk V ch. III, "Equilibrium of Normal Demand & Supply"). (Lecturer: Joachim Lund — deck "From Political Economy to (Micro)Economics.")

Theme of the module. This is where **modern (micro)economics is born**. Two things happen at once in the late 19th century. (1) **A great methodological divide** opens between the German Historical School (economics as *history* — inductive, empirical, institutional, state-friendly) and Anglo-Austrian theory (economics as *science* — deductive, abstract, market-friendly), climaxing in the **Methodenstreit** of 1883. (2) **The marginal revolution** (1871: Jevons & Menger; 1874: Walras) throws out the classical labour theory of value and replaces it with **marginal utility** — value is set by the **last unit consumed**, by the **individual at the margin**. Marshall then fuses utility (demand) with cost of production (supply) into the neoclassical synthesis — the "scissors." Economics detaches from *political economy* and from morals, and becomes the science of **utility maximization** and Homo Oeconomicus.

★ The heart of the module — from classical to neoclassical:

- **Classical political economy** (Smith → Ricardo → Mill → Marx): value is **objective** — it comes from the **cost side** (labour / cost of production); the subject is about **classes, distribution, growth and limits** (the stationary state); it is *political economy*, entangled with morals.
- **Neoclassical (micro)economics** (Jevons · Menger · Walras · Marshall): value is **subjective** — it comes from the **demand side**, from the **marginal utility** of the last unit to an individual consumer; the subject **isolates the smallest units** and the **moment of decision at the margin** (= microeconomics; macro = the sum of micro); it aspires to be an **exact science** stripped of morals, about **utility maximization**.
- **The hinge is the water-diamond paradox**: Smith could not solve it with the labour theory of value; marginalism solves it in one line — water is abundant, so its **marginal utility** (and price) is near zero, even though its *total* usefulness is huge.
- **Marshall's settlement**: value is governed by **both** utility (demand) *and* cost of production (supply) — "as reasonably dispute whether it is the upper or the under blade of a **pair of scissors** that cuts a piece of paper."

Key concepts to apply: *positivism* (Comte) *vs* *historicism* (Ranke) · *sources vs facts* · *nomothetic vs idiographic* · *deductive vs inductive* · the German Historical School (List · Roscher · Schmoller) · *institutionalism* · the Methodenstreit (Schmoller *vs* Menger, 1883) · *value in use vs value in exchange* · the water-diamond paradox · labour theory of value *vs* *subjective / marginal utility theory* · *total utility vs the final degree of utility* (marginal utility) · diminishing marginal utility · *marginalism* · the marginal revolution (Jevons · Menger · Walras) · *utility maximization* · Homo Oeconomicus · methodological individualism · Marshall's neoclassical synthesis · partial equilibrium · the scissors

· market / short-run / long-run period · money as a measuring rod · (critique) institutionalism · "bring the state back in"
· no pure competition.

The great divide: social science vs history

Core: the lecture opens on a **19th-century methodological split** that still shapes the social sciences — between a **law-seeking** science modelled on physics and a **source-interpreting** history. Economics ends up on the *science* side.

- **Two goals, two time-directions.** *Social science* knows the historical background and tries to **explain the present** (sociology, political science) **or even predict the future** (economics). *History* knows the **outcome** and tries to explain **how we got there** — it is about **transition and causation**.
- **Sources vs facts** — the deepest disagreement:
 - *Social science insists:* something **did** happen; we can read the **facts/data** in the literature; facts provide the basis for **theory building**.
 - *History insists:* we **cannot know exactly** what happened or why; we only have **sources**, which are **not facts** but material for **interpretation**.
- **The two 19th-century "isms" behind this:**

	Positivism (Comte, 1798–1857)	Historicism (Ranke, 1795–1886)
Ontology	there is a world to be discovered by scientific means	historical events must be judged on their own merits , not by our norms
Method	detect natural laws in society's evolution (data)	source criticism , a pluralism of sources, interpretation
Yields	general, quantitative data	specific, qualitative understanding

- **Why "laws of history" feel absurd — the joke that makes the point.** Henrik Nissen's spoof "**Nissen's Law**" (1981): "*whenever a state wins uncontested military hegemony in Western/Central Europe, it attacks Russia, and the following winter is unusually cold*" — Charles XII (1708), Napoleon (1812), Hitler (1941). Three data points look like a "law," but of course it is nonsense. *In plain terms:* historical regularities are not natural laws — which is exactly the historians' complaint against economic "laws."

Context: Germany vs England vs the US (the second industrial revolution)

Core: the two economic traditions grow out of **two very different national situations** during the **second industrial revolution** (railways, chemicals, electricity, steel).

- **England** — "**trapped by success**." The "**workshop of the world**," the first mover, with a global empire and a liberal, free-trade creed (**Corn Laws abolished 1846; Cobden Treaty 1860**). Complacent; the model to be explained.

- **Germany — catch-up & nation-building.** A latecomer with global ambitions, still largely agricultural, with persisting **mercantilist regulation** and a **large state bureaucracy** acting as **patron**. This is the home of **Gerschenkron's** point: the more **backward** the economy, the bigger the **role of the state** in catching up — which pushes German economists toward **protection and institutions**.
- **The US — the new giant (1860–1914).** From farming to the **largest industrial state**: the "**American system of manufacturing**" (interchangeable parts, "armory practice"), **Fordism** (assembly lines) and **Taylorism** (efficiency), railways and the telegraph, great fortunes (Morgan, Rockefeller, Carnegie), mass immigration — all behind a **weak federal state** that was **protectionist** yet **anti-trust** (Sherman Act 1890).
- **The protectionist paradox** (a lecture puzzle): *more* international competition tends to produce *more* protectionism — the loser of free competition reaches for the tariff.

Two traditions of 19th-century economic science

Core: the deck's framing — the century's economics splits into two projects asking two different questions.

	German Historical School / Historical Economics	Anglo-Saxon / Austrian classical & neoclassical theory
Question	<i>why</i> did capitalism emerge in particular countries (Britain first), and how can latecomers (Germany) catch up & survive?	<i>how</i> does capitalism function — what are its " laws "?
Emphasis	the uniqueness of national experiences; institutions	universal theory; the market mechanism
Policy lean	state intervention , protection	the forces of the market , free trade
Becomes	institutionalism	neoclassical microeconomics

The German Historical School (List · Roscher · Schmoller)

Core: an **evolutionary, dynamic** approach that **questioned the relevance of British classical doctrine** to Germany — stressing that economic institutions, history and the state differ from country to country.

- **Friedrich List — *The National System of Political Economy* (1841).** The policy voice: **free trade inside a country, protectionism on the outside** to shelter **infant industry** until it can compete. Against English individualism he stressed **collective action**: "*the association of individuals for a common end is the most efficacious mode towards the happiness of individuals.*" (Contrast Smith's invisible hand and P. Leroux's warning that English individualism turns men into "rapacious wolves.")
- **Wilhelm Roscher — *Principles of Political Economy* (1854); the "older" school.** Founded the movement (1843). The **historical method** is essential to understand the differences between

economic systems; classical theory is inappropriate to Germany's conditions. (In practice the older school's work still looked a lot like Smith or Mill.)

- **Gustav Schmoller** — *Political Economy and its Method* (1894); the "younger" school. Far more radical. **Sceptical of "laws of history"**; insisted economics must rest on **detailed empirical and historical observation** before generalizing; wanted to broaden economics toward **economic sociology**. Conservative (a Hohenzollern loyalist) **but a committed social reformer** — founder-figure of the **Verein für Sozialpolitik** ("Union for Social Policy," 1872), whose members were nicknamed "**academic socialists**" (piecemeal studies → factory laws, social insurance, working hours).

★ The Methodenstreit (1883) — the clash over method

Core: the open war that **defined the dividing line** in economics: **Gustav Schmoller** (historical/institutional) vs **Carl Menger** (theoretical/marginalist). It **split the German-speaking profession**.

Schmoller — historical / institutional	Menger — classical / marginalist
Empirical	Theoretical
Concrete	Abstract
Inductive	Deductive
Dynamic	Static
Qualitative	Quantitative
Idiographic (the specific, unique)	Nomothetic (the general law)

- **Menger's attack (1883)**. He wanted a **rigid separation**: **theoretical economics** states "**exact laws**" built on **abstractions** — **pure self-interest, omniscience, free movement**. To test such theory against messy reality is a **category mistake**: "*pure self-interest cannot exist any more than can pure oxygen.*" Two seeds that later define **Austrian economics**: methodological individualism (start from the individual, never from aggregates) and **spontaneous order**. Menger also rejected **mathematical economics**: maths shows relations between **quantities**, not the **essence** of phenomena.
 - **Schmoller** reviewed Menger scathingly; the feud turned bitter. Backhouse's verdict: the fight was **as much about policy** (Schmoller for protection, Menger against) and academic turf as about logic — the two could have agreed that **different questions need different methods**.
 - **The settlement (today's consensus)**. Sound economics needs **both**: (1) collecting **statistical information** and (2) building **analytical tools and theory** to handle it. (The lecture notes, pointedly, that **Marx** had *already* combined general static "laws" with dynamic institutional analysis in *Capital* — the subject of Lecture 8.)
-

★ The water-diamond paradox (the preconditions for marginalism)

Core: the famous puzzle that classical value theory **could not solve** — and that marginal utility **does**.

- **Smith's statement.** The word *value* has **two meanings**: **value in use** (how useful a thing is) and **value in exchange** (its purchasing power). They can come apart completely: "*Nothing is more useful than water; but it will purchase scarcely anything... A diamond, on the contrary, has scarcely any use-value; but a very great quantity of other goods may be had in exchange for it.*"
- **Smith's own answer: the labour theory of value.** Value in exchange is set by the "**toil and trouble**" of acquiring a thing — the **labour/cost** embodied in it. Diamonds are dear because they are hard to find; water is cheap because it is easy to get. This locates value on the **cost/production side** and leaves use-value hanging.
- **Why it is the hinge.** The paradox is exactly what marginalism dissolves (see below): value is not about **total** usefulness or **cost**, but about the **usefulness of the last unit** — and because water is abundant, the utility (and price) of **one more** glass is almost nothing.

Bentham & the forerunners of marginalism

Core: the raw material of marginalism — **utilitarian psychology** plus a scattering of thinkers who had already glimpsed **diminishing utility**.

- **Jeremy Bentham's utilitarianism (1748–1832).** Utility = the capacity to **increase pleasure and reduce pain**. Jevons adopts this wholesale: "*to satisfy our wants to the utmost with the least effort... to maximise pleasure, is the problem of Economics*" (Jevons, p. 37). **Pleasure and pain are the "objects of the calculus of economics."**
- **The forerunners who saw diminishing utility** (each spotted it in one corner): **Bentham** — diminishing utility **of money**; **Ricardo** — diminishing returns on **marginal lands**; **Senior** — diminishing utility **of commodities**; **Mill** — diminishing utility **of production** (marginal costs); and, in Germany, **Gossen** and, in France, **Cournot** and the engineers (Dupuit). None quite put the whole thing together.

★ The marginal revolution — what marginalism is

Core: the breakthrough of **1871–74**. Value is no longer embodied by labour; it is set at **the margin** by the **marginal utility** of the last unit — a **subjective, demand-side** theory that founds **microeconomics**.

- **The breakthrough, almost simultaneously and independently:**
 - **W. S. Jevons**, *The Theory of Political Economy* (1871) — a general value theory on utility.
 - **Carl Menger**, *Principles of Economics* (1871) — **subjective** value theory (Vienna).
 - **Léon Walras**, *Elements of Pure Economics* (1874) — value from **scarcity** ("*rareté*"), and **general equilibrium**.
 - **Alfred Marshall**, *Principles of Economics* (1890) — the **neoclassical synthesis** (supply + demand).
- **What marginalism is (deck):**

- It definitively replaces the labour theory of value with marginal utility theory.
- **Value is set at the margin:** demand depends on the **marginal utility** of the last unit, which is **subjective** and (given diminishing utility) **falls as quantity rises**. This solves the water-diamond paradox in one stroke.
- Method is **deductive**; it **presupposes free competition and complete information**.
- It **isolates the smallest units in the economy and the moment of decision at the margin** — this is **microeconomics** ("macro = the sum of micro decisions").
- It **detaches economics from "political economy"** and from morals: an **exact science** in which "morals have little to do with economic decisions." It's about **utility maximization by Homo Oeconomicus**. (It also rejects Ricardo/Mill's moral claim that land rent is "unproductive/immoral.")
- **The trio compared** (same core, three routes):

	Jevons (Britain)	Menger (Austria)	Walras (Lausanne)
Starts from	utilitarian pleasure/pain	the essence of needs (Aristotelian); not a utilitarian	scarcity (" <i>rareté</i> "); not a utilitarian
Method	mathematical (calculus)	verbal , essences & causes; anti-maths	mathematical , simultaneous equations
Signature	value = final degree of utility	subjective value; goods hierarchy; markets not in equilibrium (limited knowledge, entrepreneurs)	general equilibrium across all markets; tâtonnement ("groping") toward equilibrium

★ William Stanley Jevons (1835–1882) — value depends entirely on utility

Core: Jevons breaks with Ricardo and makes **utility** the whole basis of value, in a deliberately **mathematical** economics.

- "Economics deals with quantities, and must therefore be a mathematical science." A chemist/meteorologist by training and a methodologist (*The Principles of Science*, 1874), he was optimistic that economic quantities could be **measured** (account books, price lists, bank returns).
- His central, "somewhat novel" opinion: "*value depends entirely upon utility*." Against Ricardo (value = cost of production), Jevons says utility — specifically the benefit of the **last unit consumed** — governs value.
- The key concept: **total utility vs the final degree of utility**. The **final degree of utility** = the utility of the **last increment** of a good = what we now call **marginal utility**. The **law of the variation of utility**: the final degree of utility **diminishes** as you consume more. (So the hundredth litre of water is worth almost nothing to you.)
- Jevons's chain — how cost and value connect only *indirectly*:
 1. Cost of production determines supply;
 2. supply determines the final degree of utility;

3. the final degree of utility determines value. → Labour does not determine value directly — it only influences value by **changing supply**. The focus shifts to the individual consumer.
- **The maximization condition** (utilitarian calculus): a consumer maximizes utility when the **ratio of the marginal utilities** of two goods equals the **ratio of their prices**; a worker works until the **pain of the last hour** equals the **pleasure** its wage buys. Jevons calls this "a natural law of utility variation" and explicitly **calls for a law of supply and demand**.
 - (*Applied Jevons* — from earlier modules: **The Coal Question (1865)** and the **Jevons paradox**; and his **sunspot theory** of the trade cycle — statistical, inductive work he kept separate from the abstract theory.)*

★ Alfred Marshall (1842–1924) & neoclassical economics

Core: Marshall fuses marginal utility (demand) with cost of production (supply) into the neoclassical synthesis — the foundation of microeconomics until the 1950s. His watchword: **be realistic**.

- **The synthesis.** Marginalism (utility) explains **why people act** — but "we still need to explain the **price** of a commodity." His answer: **value is subjective and, on its own, not so important; supply and demand are equally important**. He grafts Jevons's utility theory onto his own supply-and-demand apparatus.
- **The scissors** (the module's signature quote, *Principles* Bk V ch. III): "*We might as reasonably dispute whether it is the **upper or the under blade of a pair of scissors** that cuts a piece of paper, as whether value is governed by **utility or cost of production***." Demand (utility) is one blade, supply (cost) the other; **neither alone** cuts.
- **Partial-equilibrium analysis.** To take **time** seriously, Marshall refuses Walras's all-at-once **general equilibrium** and instead studies **one market at a time** ("partial equilibrium" — the "holy cross"). Which blade dominates depends on the **period** (defined by what is free to change, not the calendar):

Period	What can adjust	Who governs price
Market period	nothing — stock is fixed	demand alone (esp. perishables like fish)
Short run	output, via unskilled labour & machinery (rising cost)	supply and demand together
Long run	everything — skilled labour, plant, methods	mainly cost of production (more output, falling cost)

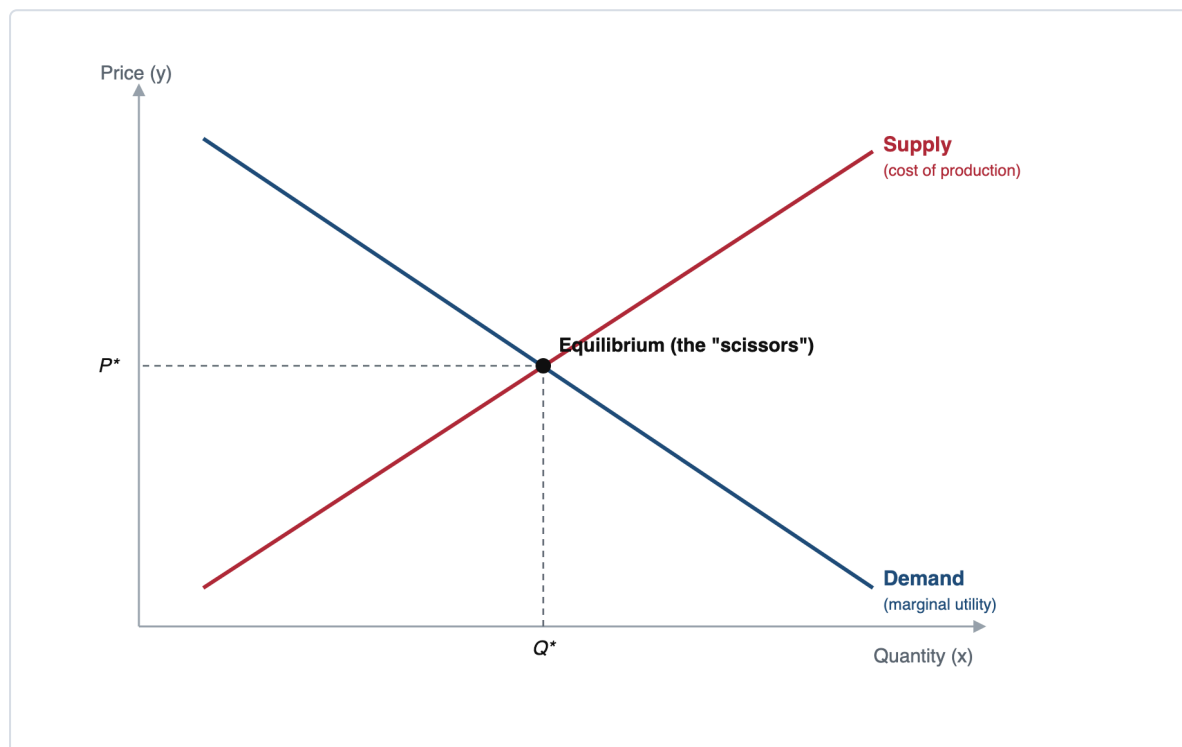
→ Rule of thumb: *the **shorter** the period, the more **demand** matters; the **longer** the period, the more **cost of production** matters.*

- **Equilibrium.** Where the **demand price equals the supply price**, output has "no tendency either to be increased or diminished" — the **equilibrium-amount** and **equilibrium-price**. A **stable equilibrium** behaves like a **pendulum** (or a stone hanging from a string): displace it, and market forces swing it back.

How the supply-and-demand curves are established (study question)

Core: build each curve from one blade of the scissors, then read price and quantity off their crossing point.

- The **demand curve** comes from **(diminishing) marginal utility**: because each extra unit is worth less, buyers will only take **more** at a **lower** price → the curve slopes **downward**.
- The **supply curve** comes from the **cost of production** (and, in the short run, **rising** marginal cost as output expands) → sellers will offer **more** only at a **higher** price → the curve slopes **upward**.
- **Equilibrium** is where they **cross** — the **scissors** — fixing the **equilibrium price P^*** and **quantity Q^*** . (Marshall's convention, still used: **price on the y-axis**, **quantity on the x-axis**.)



How to read it (illustrative diagram). The **blue** line is **demand** — low marginal utility of extra units means buyers only take more as the price falls. The **red** line is **supply** — rising production costs mean sellers only offer more as the price rises. They meet at the **equilibrium (P^* , Q^*)**: the one price at which the amount people want to buy equals the amount firms want to sell. Move the price above P^* and supply exceeds demand (it falls back); below P^* , demand exceeds supply (it rises) — the pendulum returns to the cross.

Economics as a science — and the measuring rod of money

- **Marshall's definition:** economics is "a study of mankind in the ordinary business of life" — needs, wishes, incentives, motives, choices (so, partly **psychology**).
- **How to measure the unmeasurable? Money.** The **price we are willing to pay** for something expresses our **individual utility** at the margin — and since money is an **exact scale**, Marshall concludes **economics can be the most exact of the social sciences** (compare Jevons: "economics deals with quantities, so it must be mathematical"). But he is cautious: "**economics is less exact than the natural sciences**, but progress is being made towards greater precision."

- **The evolutionary temper.** Marshall preferred **biological** to **mechanical** metaphors — motto "*Natura non facit saltum*" ("nature makes no jumps": change is **gradual, continuous**). He saw **firms** passing through a **life cycle** (young & vigorous → old & displaced): *an industry is like a forest — the same even as every tree changes*. He also noted **external economies** (economies of scale) and the realities of **monopoly and duopoly**. Tellingly, he buried his **diagrams in footnotes** and **algebra in an appendix** — he wanted the book readable by **businessmen**.

★ What is "classical" and what is "neo-" in neoclassical economics? (study question)

Core: "neoclassical" = **marginalism**, *plus* what survived of the classics. Stanley Brue's definition: "*basically marginalism with a judicious recognition of the surviving contributions of the classical school*" (individual freedom of enterprise, private property, the market mechanism).

	Classical (Smith · Malthus · Ricardo · Mill · Marx)	Neoclassical (Jevons · Menger · Walras · Marshall)
Theory of value	objective — labour / cost of production	subjective — marginal utility
Side of the market	supply / production	demand / the consumer
Unit of analysis	classes (landlords, capitalists, workers) & distribution	the individual at the margin (micro)
Big question	growth and its limits (the stationary state)	price & allocation via equilibrium
Human model & morals	<i>political</i> economy, entangled with ethics	Homo Oeconomicus , utility maximization, "value-free"

- **Marshall is the "neo-":** he keeps the classical **supply/cost** blade and adds the new **utility/demand** blade — hence *neo-classical*. (Transitional figures **Mill** and **Marx** keep the labour theory of value → they count as **classical**, even while adding new ideas on distribution and crises.)

Critique of neoclassical economics

Core: the assumptions that make the theory elegant are also its weak points — the charges that institutionalists and later Keynes press.

- **History matters** — the early **institutionalist** reply: abstract universal "laws" ignore how institutions and context actually shape economies.
- **"Bring the state back in"** — the state is not a distortion but the **protector of capitalism** (Gerschenkron, the historical school).
- **It cannot explain growth or crises** — the model tends to **static equilibrium**; **Keynes** will argue economies can sit in an **unemployment disequilibrium** that the market does not self-correct.
- **No empirical evidence** — **only assumptions**. Above all, **"pure competition" does not exist**: real markets tend toward **capital concentration, cartels and monopoly** — exactly what Marshall half-

admitted with external economies and De Beers.

Cases & examples

- **"Nissen's Law"** (1981 spoof) — Charles XII (1708), Napoleon (1812), Hitler (1941): why three data points are not a law of history.
- **Smith's water & diamonds** — the paradox of value in use vs value in exchange.
- **List's infant industry** — free trade inside, protection outside (the German catch-up programme).
- **The Methodenstreit, 1883** — Schmoller vs Menger as the emblem of the history-vs-theory split.
- **Marshall's "pair of scissors"** and the **fish market** (perishables → price set by demand) vs **wheat** (storable → sellers hold out) — the period analysis.
- **De Beers** (monopoly) / **Apple & Microsoft** (duopoly) — Marshall's own reality checks against "pure competition."

Key terms

Term	Meaning
Positivism / Historicism	Comte: society has discoverable laws / Ranke: judge the past on its own terms, via sources
Nomothetic / Idiographic	seeking general laws / describing the specific and unique
Deductive / Inductive	reasoning from theory to cases / generalizing from observed facts
German Historical School	economics as history & institutions; state-friendly (List, Roscher, Schmoller)
Methodenstreit (1883)	the Schmoller–Menger "struggle over method" that split German economics
Value in use / value in exchange	usefulness / purchasing power (the two senses of "value")
Water-diamond paradox	the useful thing (water) is cheap, the useless one (diamond) dear
Labour theory of value	value comes from the labour/cost embodied in a good (classical)
Marginal utility / final degree of utility	the utility of the last unit consumed (Jevons's term)
Diminishing marginal utility	each extra unit is worth less than the one before
Marginalism	value is set at the margin by marginal utility, not by labour
Homo Oeconomicus	the rational, self-interested utility-maximizing individual
Methodological individualism	all analysis must start from the individual
Neoclassical synthesis	Marshall's fusion of marginal utility (demand) + cost of production (supply)
Partial equilibrium	analysing one market at a time (to handle time), not all at once
The scissors	value is set by both blades — utility and cost of production
Equilibrium price / amount	where demand price = supply price; no tendency to change
Market / short-run / long-run period	nothing adjusts → only output adjusts → everything adjusts
Tâtonnement	Walras's "groping" trial-and-error toward general equilibrium

★ Study-question answers (Joachim)

1. **What is the water-diamond paradox?** Smith's puzzle that **value in use and value in exchange come apart**: water is supremely useful yet almost free, while a diamond is nearly useless yet hugely expensive. Classical **labour value** "explained" it by cost (diamonds are hard to get); **marginalism** solves it properly — price follows **marginal utility**, and because water is **abundant**, the utility (and price) of *one more* unit is near zero. See the paradox.
2. **What is "classical" and what is "neo-" in neoclassical economics?** The "neo-" is **marginalism** — subjective **marginal-utility** value, the individual consumer, equilibrium, micro, Homo Oeconomicus.

The "classical" that survives is **supply/cost of production**, private property, free enterprise and the market. Marshall's "neo-classical" economics **keeps the classical cost blade and adds the utility blade** (Brue: "marginalism with a judicious recognition of the surviving contributions of the classical school"). See [classical vs neo-](#).

3. **What is marginalism?** The doctrine that **economic value is determined at the margin** — by the **marginal (final-degree) utility of the last unit**, which **diminishes** as quantity rises — **not** by labour or cost. It is **subjective, deductive**, assumes **free competition and full information**, isolates the **individual's decision at the margin** (hence *microeconomics*), and reduces economics to **utility maximization**. It arrives in **1871** (Jevons, Menger) and **1874** (Walras). See [marginalism](#).
4. **What were Jevons's main contributions? And Marshall's?** **Jevons:** declared "**value depends entirely upon utility**", introduced the **final degree of utility** (= marginal utility) and the **law of its diminution**, replaced the [labour theory of value](#) with the chain **cost → supply → final degree of utility → value**, insisted economics is a **mathematical** science of pleasure/pain (Bentham), and shifted the focus to the **individual consumer**. **Marshall:** the [neoclassical synthesis](#) — **supply and demand** jointly set price (the **scissors**), **partial-equilibrium** and **period** analysis to handle time, the modern **supply-demand diagram** and **equilibrium price**, **money as a measuring rod** for utility, and the textbook (*Principles*, 1890) that founded **microeconomics** to the 1950s. See [Jevons](#) · [Marshall](#).
5. **How are the supply-demand curves established?** The **demand curve** is built from [diminishing marginal utility](#) (buyers take more only as price falls → downward slope); the **supply curve** from **cost of production / rising marginal cost** (sellers offer more only as price rises → upward slope); **price and quantity** are fixed where they **cross** — Marshall's **scissors** — at the stable **equilibrium** (P^* , Q^*), with **price on the y-axis and quantity on the x-axis**. See [how the curves are established](#).

★ Fun facts & memorable details (Module 7: Marginalism & Neoclassical Economics)

Sticky bits from the "political economy → (micro)economics" lecture.

- [The scissors that settled a century-old fight](#). Marshall ended the "utility vs labour" value war with one image: asking whether utility or cost sets value is like asking **which blade of the scissors cuts the paper**. Both do.
- "**Pure self-interest cannot exist any more than can pure oxygen**." Menger's jab (1883) defending abstract theory — economic "laws" are built on idealisations, so you cannot refute them by pointing at messy reality.
- **1871, twice**. The marginal revolution's two founding books — **Jevons's** *Theory of Political Economy* and **Menger's** *Principles* — appeared in the **same year**, independently, in different countries and languages. Walras joined in 1874; the three only discovered each other in the **mid-1870s**.
- **The socialist marginalist**. **Walras** called himself a **socialist** on the strength of his plan to **tax rising land rents** — using the same marginalist toolkit his Austrian admirers would wield against socialism.
- **Nissen's Law**. A Danish historian's deadpan "law" — win hegemony in Europe, invade Russia, freeze — skewering the idea that history obeys natural laws (Charles XII, Napoleon, Hitler).

- **Marshall hid the maths.** Trained as a mathematician, he still buried his **diagrams in footnotes and algebra in an appendix** so **businessmen** could read the *Principles* — and refused to publish a growth model because he wasn't sure about the equations.
 - **"Nature makes no jumps."** Marshall's motto *Natura non facit saltum* — change is gradual; an **industry is like a forest**, the same even as every tree is born, grows and dies.
 - **Economics loses its first name.** By 1900 "**political economy**" was giving way to "**economics**" — the new science had detached itself from politics *and* from morals: now it was about **utility maximization** and **Homo Oeconomicus**.
-